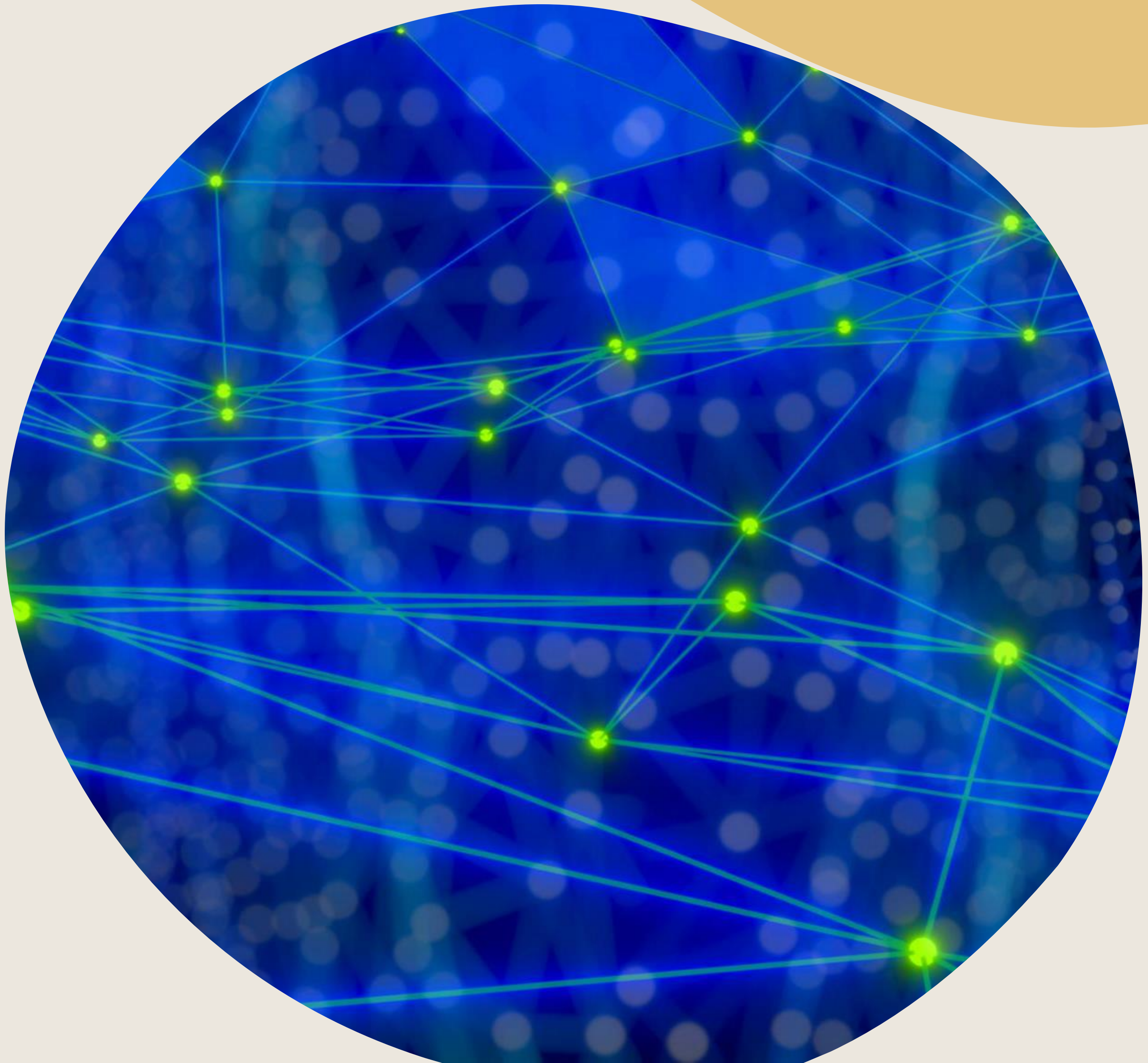
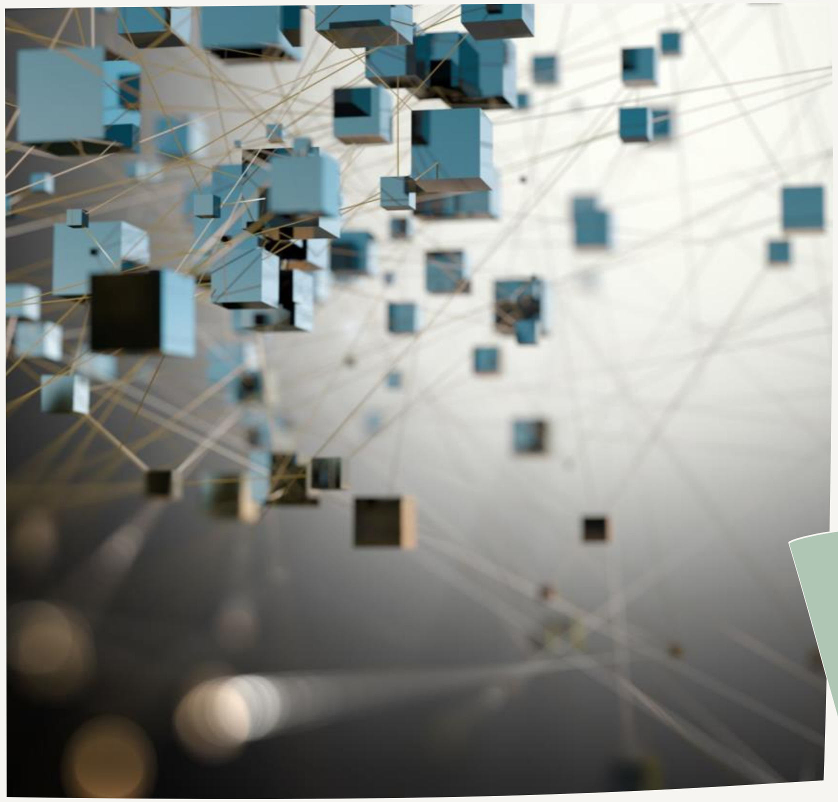
Agent-Based Modeling in Python with Mesa

Simulating complex systems using Python programming framework



Introduction to Agent-Based Modeling

Agent-Based Modeling (ABM) Overview



Modeling Individual Agents

ABM models heterogeneous agents with decentralized decision-making and local interactions.

Emergent Macroeconomic Phenomena

Simple rules followed by agents lead to complex emergent behavior not captured by traditional models.

Applications in Economics

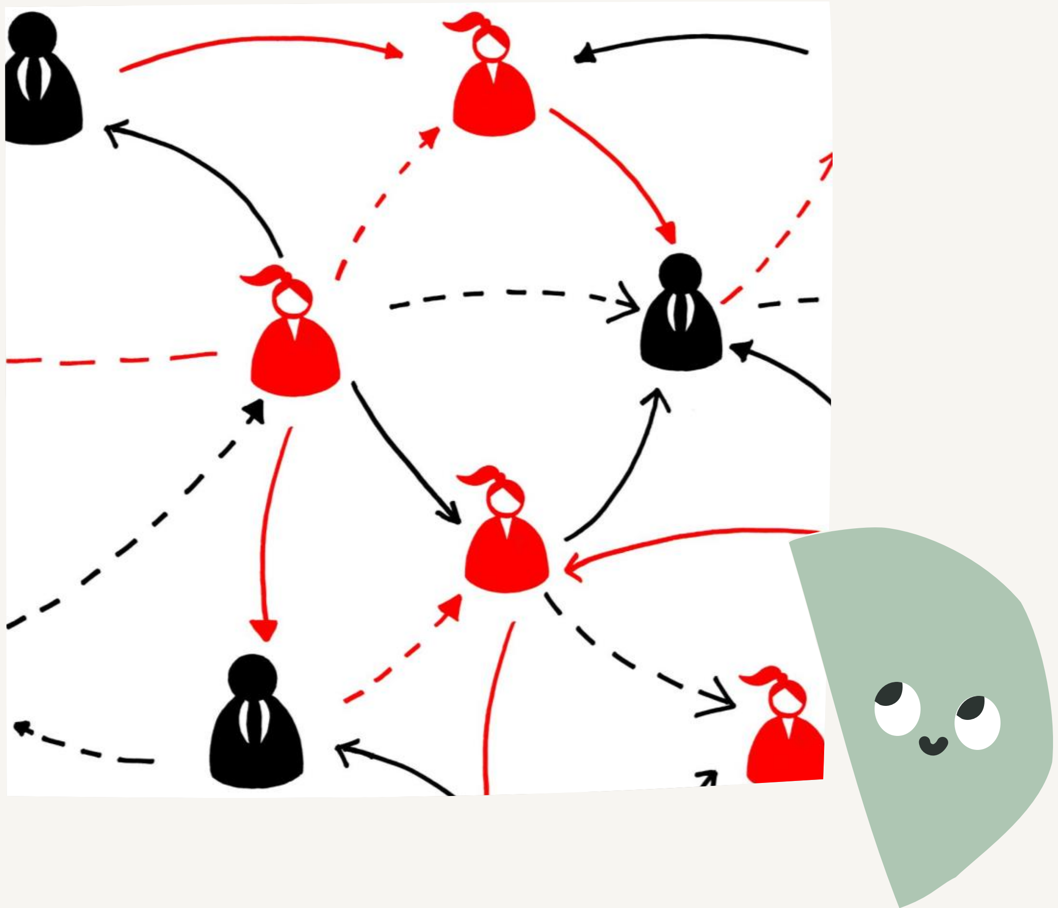
ABM studies market dynamics, inequality, contagion, and policy impacts where analytical solutions are limited.

Insights into Dynamics

ABM enables observation of transitional dynamics, path dependence, and non-equilibrium system behavior.

Why Agent-Based Modeling in Economics

Economic Motivation for ABM



Limitations of Traditional Models

Traditional economic models often assume homogeneity and perfect rationality, limiting real-world applicability.

Heterogeneous Agent Representation

ABM allows modeling of diverse agents with bounded rationality interacting locally in a structured environment.

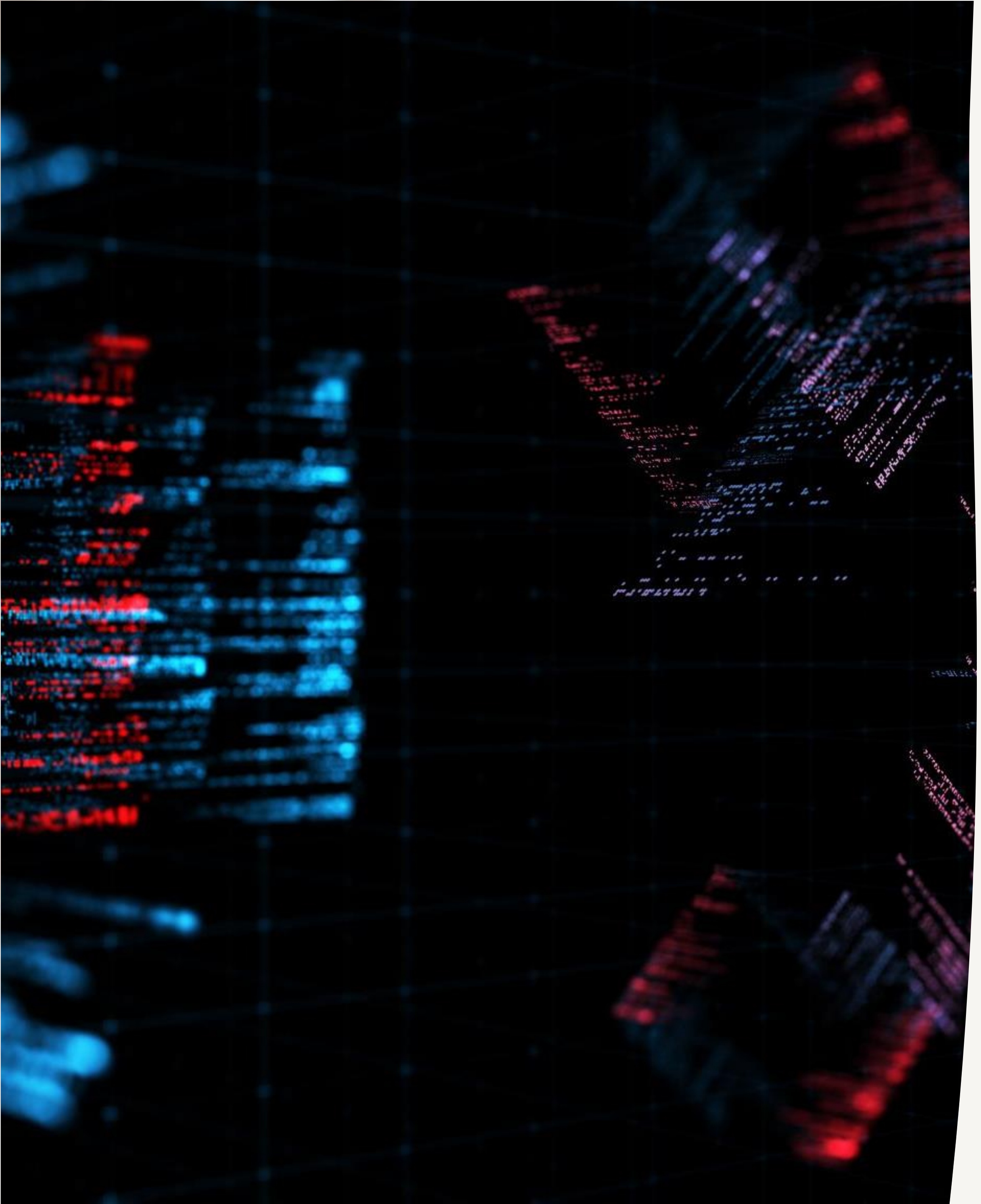
Applications and Benefits

ABM is valuable for exploring policy scenarios, market design, and analyzing complex social interactions.

Educational Value for Students

ABM strengthens computational thinking and complements equilibrium-based economic theory for graduate students.

What Is Mesa?



Mesa: A Python Framework for ABM

Open-source Python Library

Mesa is an open-source Python library designed for creating and visualizing agent-based models easily.

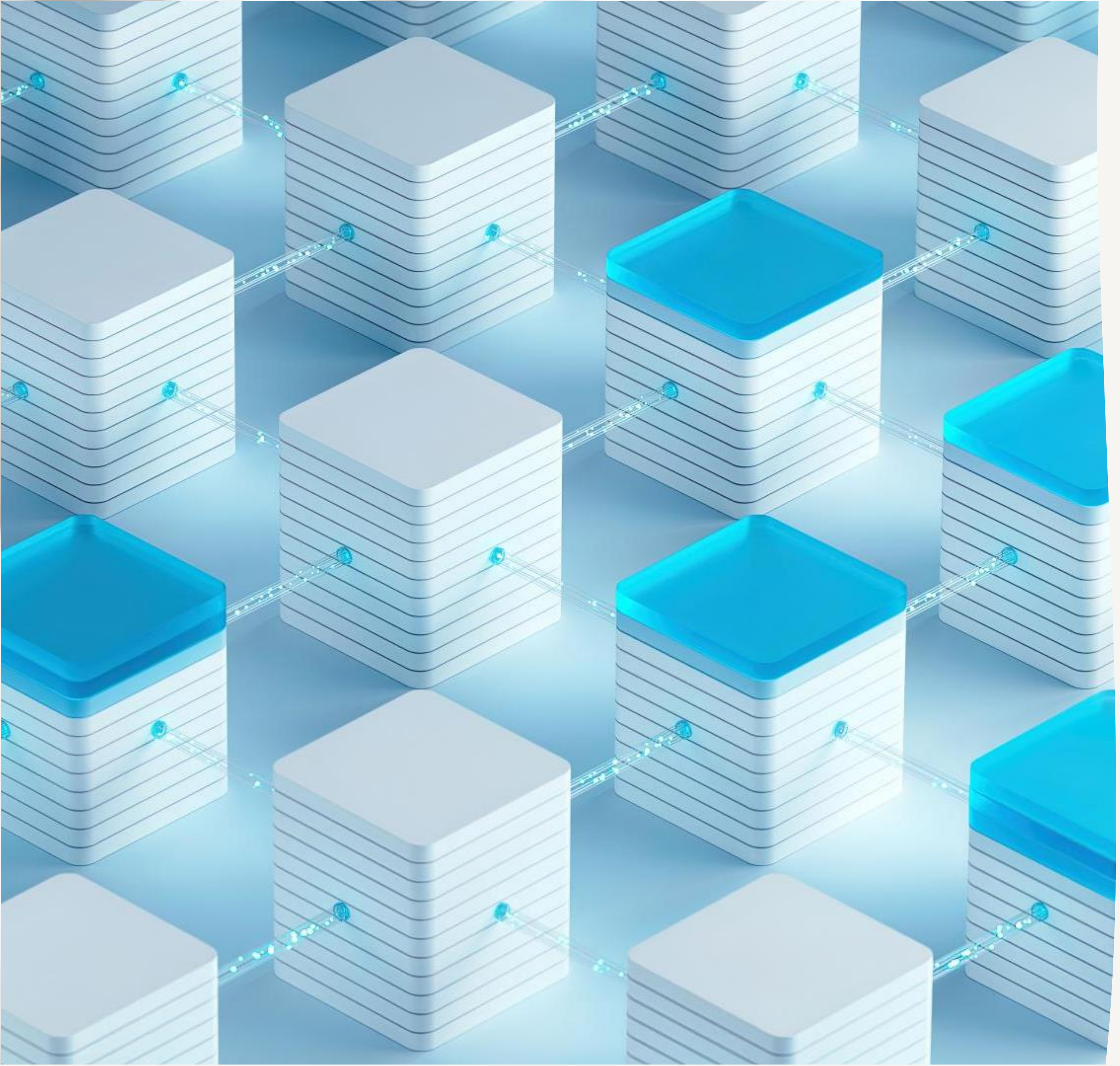
Integration with Scientific Libraries

Mesa integrates smoothly with libraries like NumPy, Pandas, and Jupyter for data-driven analysis.

User-friendly and Modular Design

Mesa uses standard Python syntax and modular design, lowering barriers for users and encouraging experimentation.

Design Goals and Architecture



Design Philosophy of Mesa

Core Design Principles

Mesa focuses on accessibility, modularity, and extensibility to support diverse agent-based modeling needs.

Separation of Concerns

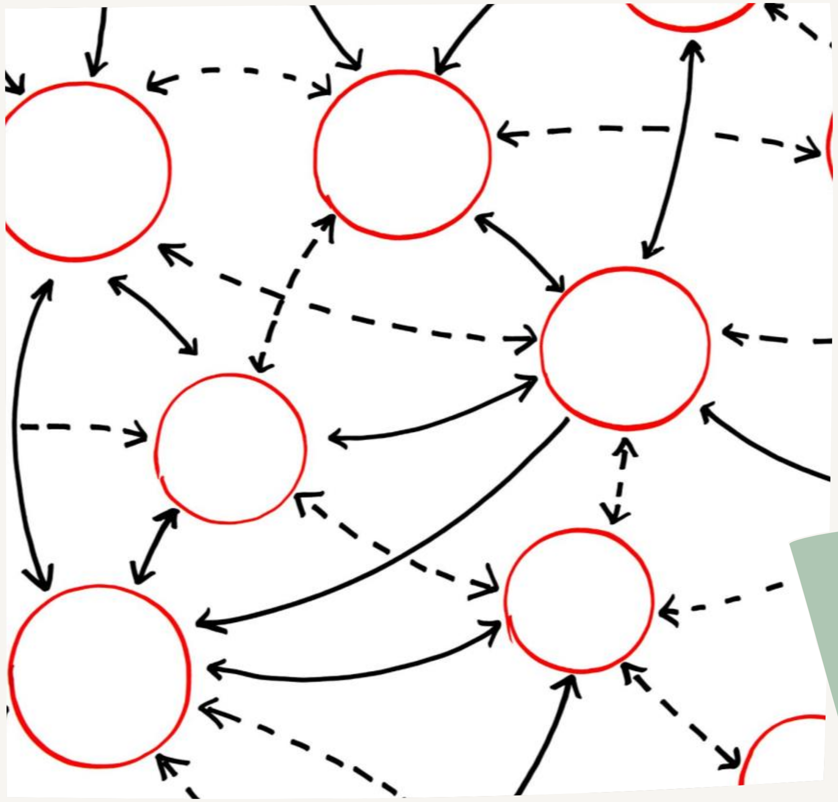
Distinct components for agent behavior, scheduling, space, data collection, and visualization enhance clarity and reuse.

Educational Focus

Mesa's architecture simplifies complex simulations, promoting transparency and reproducibility for teaching and research.

Core Components of Mesa

Model and Agent Classes



Model Class Overview

The Model class encapsulates the entire simulation system including global parameters and scheduling.

Agent Class Functionality

Agents represent individual decision units with their own state and behavior encoded in a step method.

Micro-Macro Economic Structure

The micro–macro design mirrors economic reasoning, linking individual actions to macroeconomic outcomes.

Customization via Subclassing

Students can subclass Model and Agent classes to create custom economic agents and maintain clarity.

Scheduling, Space, and Data Collection

Agent Scheduling Control

Schedulers manage the timing and order of agent actions, enabling study of asynchronous and stochastic behaviors.

Structured Interaction Spaces

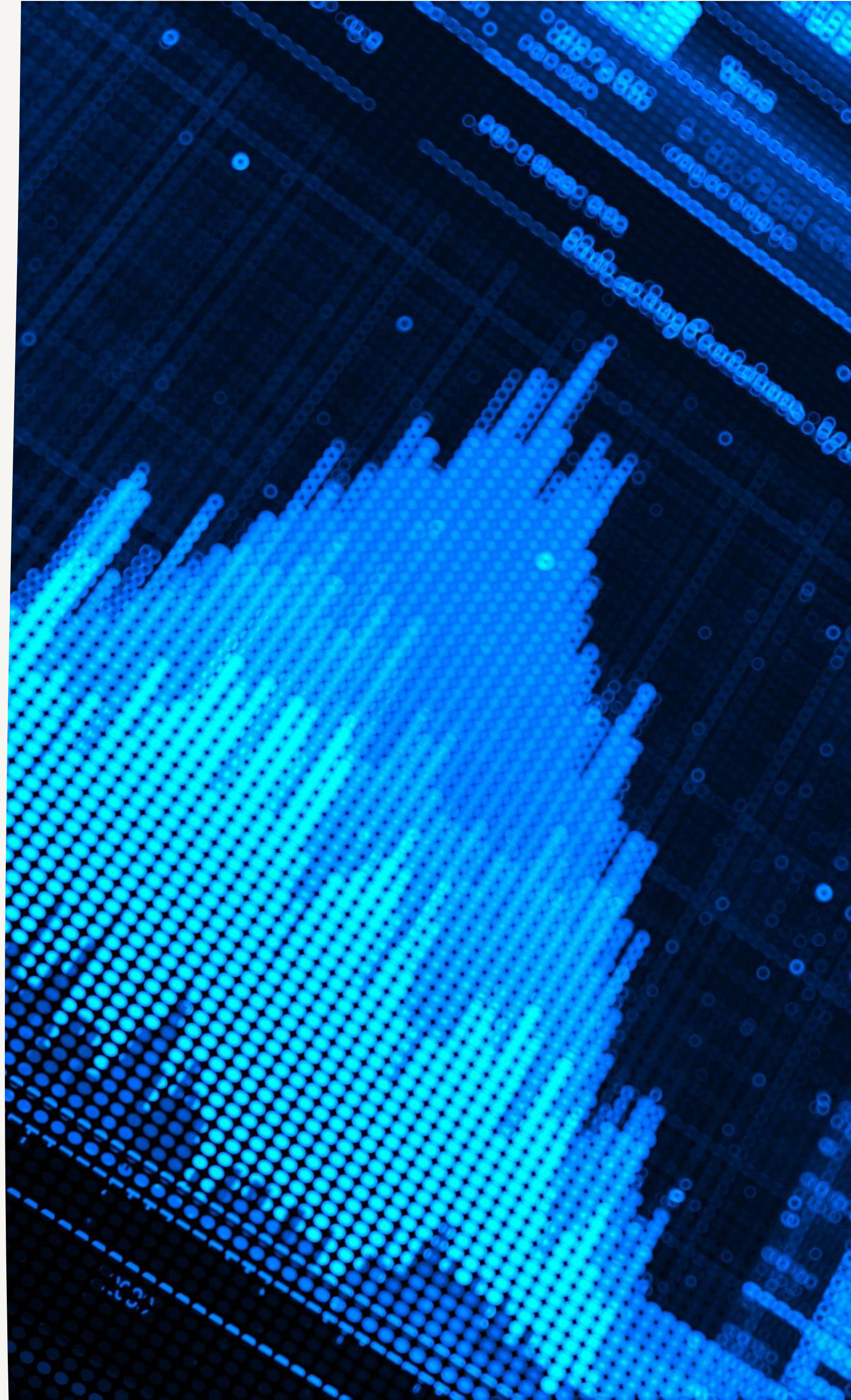
Grids and networks define agent interactions, emphasizing the role of topology in economic systems.

Integrated Data Collection

Built-in DataCollector records variables during simulations, compatible with Pandas for analysis and visualization.

Minimal Python Code Examples

Running Simulations and Collecting Results



Simulation Execution

Running simulations steps the model forward in time, allowing observation of evolving system states.

Data Collection

DataCollector class records aggregate variables like wealth and inequality at each simulation step.

Data Analysis

Simulation results output as Pandas DataFrames enabling econometric and visualization analysis.

Learning and Intuition

Experimenting with parameters helps students understand system dynamics and policy sensitivity.

Why Mesa for an Economics Course

Pedagogical Advantages

Alignment with Student Skills

Mesa leverages students' existing Python knowledge, facilitating knowledge transfer across courses and research.

Clear Model Specification

Mesa promotes explicit assumptions and systematic experimentation, essential for rigorous economic research.

Bridging Theory and Practice

Mesa helps students integrate theory, simulation, and data analysis to engage with complex economic systems.

Fostering Critical Thinking

Mesa encourages curiosity, experimentation, and critical evaluation of model assumptions and results.